



UTM

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SECJ3553 ARTIFICIAL INTELLIGENCE - SECTION 01

PROJECT ASSIGNMENT 1

THEME: HEALTHCARE

(MINDCARE AI - INTELLIGENT MENTAL HEALTH SUPPORT CHATBOT)

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Knowledge Representation

Knowledge representation (KR) in the MINDCARE AI Mental Health Support Chatbot defines how the system logically models user symptoms, mental conditions and safety risks to perform intelligent reasoning. For this AI solution, the selected Knowledge Representation method is First-Order Logic (FOL).

The system uses a Knowledge Base (KB) containing user-reported symptoms and a set of logical rules that guide the chatbot's decision-making process.

1. Identification of Depression

Description:

If the user reports both persistent sadness and a loss of interest (anhedonia), the system infers that the user has signs of Depression.

Explanation:

When a user mentions experiencing persistent sadness and a loss of interest in daily activities, the chatbot interprets these inputs as key indicators of Depression. This rule helps to ensure that the system can identify depressive symptoms even when the user does not explicitly mention the word “depression”. This KR supports the goal of early identification of mental health issues to provide the user with appropriate emotional support.

Knowledge Representation:

IF Persistent_Sadness = TRUE AND Loss_Interest = TRUE, THEN Depression = TRUE

First Order Logic:

$\forall x((\text{HasSymptom}(x, \text{PersistentSadness}) \wedge \text{HasSymptom}(x, \text{LossOfInterest})) \rightarrow \text{HasCondition}(x, \text{Depression}))$

2. Identification of Anxiety

Description:

If the user reports either a racing heart or constant worrying, the system infers that the user has signs of Anxiety.

Explanation:

Anxiety can present as continuous worrying or physical symptoms such as racing heart. When the user describes either of these conditions, the chatbot identifies the user as experiencing anxiety. This KR supports the goal of recognizing anxiety symptoms accurately even when the user only mentions one of the common indicators.

Knowledge Representation:

IF RacingHeart = TRUE $\vee$ ConstantWorry = TRUE, THEN anxiety = TRUE

First Order Logic:

$\forall x((\text{HasSymptom}(x, \text{RacingHeart}) \vee \text{HasSymptom}(x, \text{ConstantWorry})) \rightarrow \text{HasCondition}(x, \text{Anxiety}))$

3. Safety Risk Assessment**Description:**

Safety is the priority. If a user mentions self-harm or suicidal thoughts, the system immediately classifies them as "High Risk."

Explanation:

User safety is the system's highest priority. When the chatbot detects statements related to self-harm or suicidal thoughts, it immediately categorizes the user as High Risk. This prevents casual responses and ensures that the system moves directly into emergency guidance mode. This KR supports the goal of ensuring user safety through immediate crisis detection.

Knowledge Representation:

IF SelfHarm = TRUE OR Suicidal_Thought = TRUE, THEN HighRisk = TRUE

First Order Logic:

$\forall x((\text{HasSymptom}(x, \text{SelfHarm}) \vee \text{HasSymptom}(x, \text{SuicidalThoughts})) \rightarrow \text{IsHighRisk}(x))$

4. Mild Anxiety Recommendation**Description:**

If the user is identified as having Anxiety but is not classified as High Risk, the chatbot recommends a calming breathing exercise.

Explanation:

When the user is experiencing anxiety but does not exhibit any high-risk symptoms, the chatbot provides a safe and calming intervention such as a breathing exercise. This KR supports the goal of offering practical coping strategies that help users manage mild emotional distress.

Knowledge Representation:

IF Anxiety = TRUE AND HighRisk = FALSE, THEN BreathingExercise = TRUE

First Order Logic:

$\forall x((\text{HasCondition}(x, \text{Anxiety}) \wedge \neg \text{IsHighRisk}(x)) \rightarrow \text{Recommend}(x, \text{BreathingExercise}))$

5. Emergency Intervention

Description:

If the user is classified as High Risk (regardless of other conditions), the system must recommend contacting a crisis hotline immediately.

Explanation:

If the user is classified as high risk, the chatbot must immediately provide emergency resources. This ensures that individuals experiencing severe emotional distress receive appropriate guidance. This KR supports the goal of providing life-saving information during critical situations by ensuring that emergency responses override all other system actions.

Knowledge Representation:

IF HighRisk = FALSE, THEN ContactCrisisHotline = TRUE

First Order Logic:

$\forall x(\text{IsHighRisk}(x) \rightarrow \text{Recommend}(x, \text{ContactCrisisHotline}))$

Knowledge Representation Involved to Achieve Goals

The MindCare system aims to provide safe, supportive, and educational mental health guidance. Using First-Order Logic (FOL) ensures that its decisions are structured, explainable, and predictable, which is essential in the mental health domain.

Goal 1: Deterministic Safety

Related KRs:

KR3: High-Risk Detection

KR5: Emergency Intervention

Explanation:

These primary outcomes ensure that whenever any mention of self-harm or suicidal thoughts is made, the high-risk safety measure is activated immediately. The system removes ambiguity and has a predictable safety response by using strictly defined first-order logic (FOL) rules of implication. In turn, the given approach serves the main goal of protecting users, as the life-threatening signs will always trigger emergency actions in case of such indicators.

Goal 2: Transparent Inference

Related KRs:

KR1: Identification of Depression

KR2: Identification of Anxiety

Explanation:

These major findings combine user-reported symptoms using FOL connective to conclude about the underlying psychiatric conditions. The system provides a clear structure of the causality between symptoms and the diagnoses that can easily be explained and audited by explicitly defining the causal relationship between symptoms and diagnoses. Such a methodological transparency helps in the overall aim of making diagnostic logic explicit, structured and understandable.

Goal 3: Consistency across interaction

Related KR:

KR2: Identification of Anxiety

KR4: Recommendation for Mild Anxiety

KR6: Stress Management Recommendation

Explanation:

Such regulations act in a uniform manner based on a unified logical platform whenever a user interacts with the chatbot. The system provides stable and predictable reactions to indications of anxiety through the use of predicate-based modeling, i.e. mapping between indications of anxiety and the execution of breathing exercises, irrespective of the temporal background or descriptive variation advanced by users. This fit keeps the goal of maintaining quality user interactions.

Goal 4: Explainable Recommendation

Related KRs:

KR4: Recommendation for Mild Anxiety

KR5: Emergency Intervention

KR7: Mood Tracking Suggestion

Explanation:

Each of the recommendations provided by the system is based directly on a given FOL rule, and thus, the reasoning behind it can be explained completely. Even in the case of whether the recommendation is associated with the methods of grounding, or whether it concerns the provision of a crisis hotline, every choice can be explained by transparent reason. This form of evidence is the basis of the intention to develop trust by making decision-making processes clear and rule-based.

Goal 5: Support the system's educational and self-help goal

Related KRs:

KR6: Stress Management Recommendation

KR7: Mood Tracking Suggestion

Explanation:

These are rules that essentially provide you with some simple self-help procedures, such as grounding exercises and a mood tracking journal that monitors trends in your symptoms or behavior. The system achieves its objective of imparting mental-health knowledge and useful tools to the user by taking a systematic KR approach to leading you to a more successful emotional awareness and coping strategies.